Advanced Mini DJ Controller

An interactive electronic musical system combining Arduino hardware, Processing software, real-time audio manipulation, and a professional CDJ-style visual interface.

MARK BONETE · IKER MARTINEZ · MARCEL FARELO

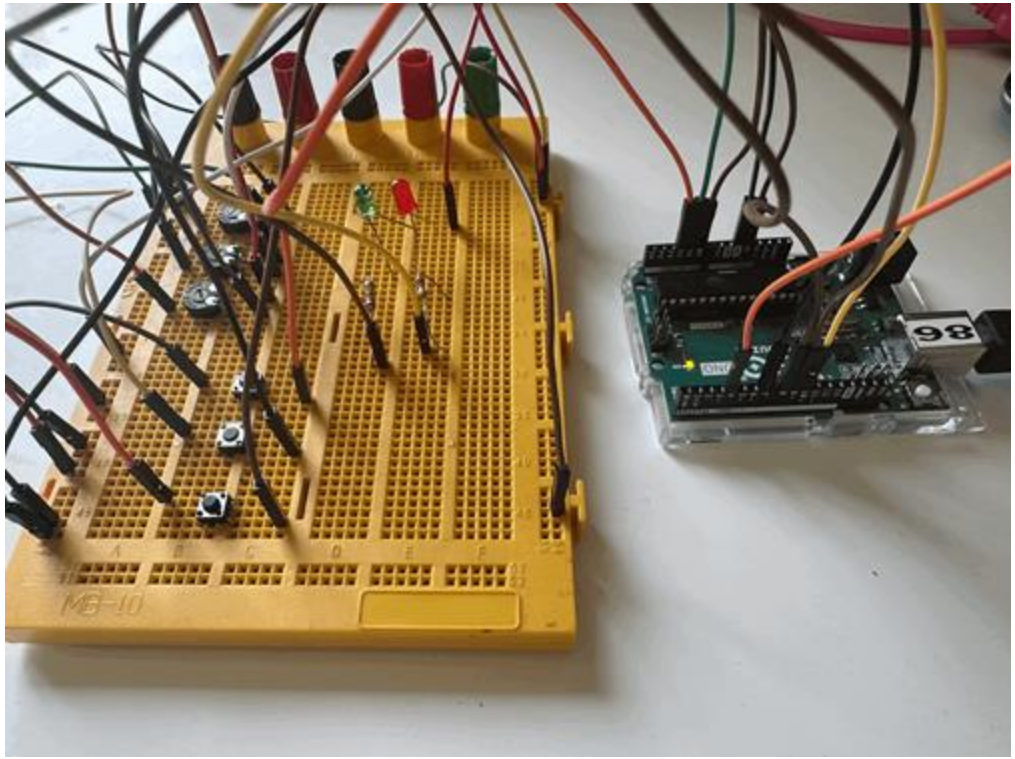


Project Overview

This project connects a physical Arduino controller — potentiometers, buttons, and LEDs — to a Processing-based audio and visual interface. Users control playback in real time while Processing renders a professional CDJ-style display with virtual decks, jog wheels, waveform visualisation, and VU meters. The result is a functional interactive prototype that behaves like a simplified DJ controller.



Hardware Architecture



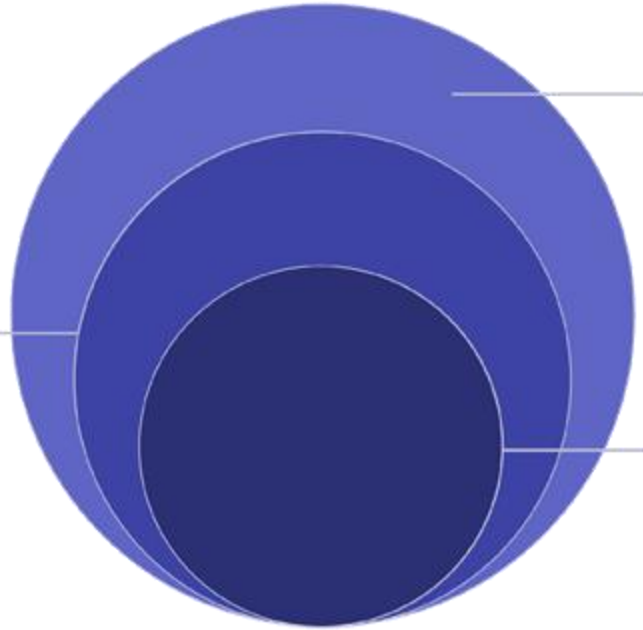
Components

- Arduino running StandardFirmata
- 3 potentiometers (A0, A1, A2)
- 3 push buttons (D4, D7, D8)
- 2 LEDs (D9, D10)
- Breadboard and jumper wires

Control Mapping

- A0 — Volume
- A1 — Playback speed / pitch
- A2 — DJ Fade / Pump FX
- D4 — Play / Stop · D7 — Next Song
- D8 — CUE / Restart
- D9 — Playback LED · D10 — CUE LED

Software Architecture



Engine: Audio & Visual

Sound lib, FFT, CDJ interface rendering

Bridge: Processing

Serial comms, port detection, smoothing

Core: Firmata I/O

Reads pins, writes LED states

Processing communicates directly with Arduino via StandardFirmata, eliminating the need for a custom serial protocol. The software reads analog and digital inputs, controls LEDs, manages two virtual decks, applies audio effects, and renders the full CDJ-style interface with real-time FFT visualisation.



Dual Deck Audio System

Deck A & Deck B

Two virtual decks load different tracks. Only one deck plays at a time, preventing audio overlap and ensuring system stability.

Deck Switching

Pressing Next Song loads a new track into the inactive deck, then swaps the active deck seamlessly.

CUE Function

Button 8 restarts the active deck to its starting point — mirroring a real CDJ controller workflow.

Visual Interface Design



Interface Elements

- Two large virtual CDJ-style decks
- Rotating jog wheels during playback
- Track information screens
- Central mixer with virtual knobs and sliders
- VU meters and active deck highlighting
- Real-time FFT waveform reacting to live audio

The interface was redesigned to closely resemble a professional CDJ setup. The waveform evolved from a simple visual into a dynamic FFT spectrum that responds to the actual sound being played.

DJ Fade / Pump FX

Controlled by potentiometer A2, this effect was selected after testing filters, gates, and echoes. It is the most audible and demo-friendly option.

1

A2 Low

Audio plays normally — no effect applied

2

A2 Mid

Subtle rhythmic volume fade — gentle pulse

3

A2 High

Strong pump effect — clear build-up and drop



Objectives vs Final Results

Basic Objectives	Status
Read analog/digital inputs	✅ Completed
Stable Arduino–Processing comms	✅ Completed (StandardFirmata)
Visual feedback	✅ Exceeded (full CDJ GUI)
Intermediate Objectives	**Status**
Map hardware to visual parameters	✅ Completed
Multi-control interface	✅ Completed
Audio playback control	✅ Expanded (dual deck)
Advanced Objectives	**Status**
Volume, speed, track control	✅ Completed
LED feedback	✅ Completed
DJ-style interface & waveform	✅ Exceeded (FFT waveform)

Challenges & Solutions

Audio Files Not Found

Files placed in the Processing data folder; filenames verified.

Wrong Serial Port (macOS)

Auto-detection added — searches for `usbmodem` / `usbserial` ports.

Audio Overlap on Track Switch

System now calls `stop()` before starting a new track; only one deck plays at a time.

Speed Potentiometer Instability

Smoothing added via `lerp()` and a noise threshold.

Effects Not Audible Enough

Replaced with the more noticeable DJ Fade / Pump FX.

Interface Elements Overlapping

Mixer layout reorganised for a cleaner, more professional appearance.

Conclusion & Future Directions

The Advanced Mini DJ Controller successfully integrates Arduino hardware, Processing software, real-time audio control, and a professional visual interface — evolving from a basic prototype into a fully functional dual-deck DJ system.

What We Built

Dual-deck controller with physical controls, LED feedback, CDJ-style interface, and real-time FFT waveform.

Future Improvements

Simultaneous two-track mixing, crossfader, advanced effects (echo, reverb), BPM sync, RGB LEDs, and a physical enclosure.

